

Types of Numbers

Simplified Definitions with Examples & Notes

In maths, numbers are grouped into different families. Each family is built on top of the previous one. This guide explains each type in simple words with examples and quick revision notes.

1. Natural Numbers (N)

Definition: Counting numbers that start from 1 and go on forever: 1, 2, 3, 4, ...

Symbol: N

Examples: 1, 2, 3, 4, 5, ... (e.g. number of students in a class)

Notes:

- The smallest natural number is 1.
- Zero (0) is NOT a natural number.
- They are always positive whole numbers.

2. Whole Numbers (W)

Definition: All natural numbers together with 0: 0, 1, 2, 3, 4, ...

Symbol: W

Examples: 0, 1, 2, 3, 4, ...

Notes:

- Whole numbers = Natural numbers + 0.
- The smallest whole number is 0.
- Every natural number is a whole number (but 0 is whole, not natural).

3. Integers (Z)

Definition: All whole numbers plus their negatives: ..., -3, -2, -1, 0, 1, 2, 3, ...

Symbol: Z

Examples: -100, -5, -1, 0, 3, 47

Notes:

- Includes positive numbers, negative numbers and zero.
- There are no fractions or decimals in integers.
- Zero is an integer that is neither positive nor negative.

4. Rational Numbers (Q)

Definition: Any number that can be written as a fraction p/q , where p and q are integers and q is not equal to 0.

Symbol: Q

Examples: $1/2$, $-3/4$, 5 ($= 5/1$), 0.75, $0.333... (= 1/3)$

Notes:

- Decimals that terminate (end) or repeat are rational.
- Every integer is also a rational number.
- The denominator q can never be 0.

5. Irrational Numbers

Definition Numbers that CANNOT be written as a simple fraction p/q . Their decimal part goes on forever without repeating.

Examples: $\pi = 3.14159\dots$, $\sqrt{2} = 1.41421\dots$, $e = 2.71828\dots$

Notes:

- They are non-terminating and non-repeating decimals.
- Square roots of non-perfect squares are irrational ($\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$).
- π and e are the most famous irrational numbers.

6. Real Numbers (R)

Definition All rational and irrational numbers together. Any number that can be placed on the number line is a real number.

Symbol: R

Examples: -2, 0, $1/2$, 7.5, $\sqrt{2}$, π

Notes:

- Real numbers = Rational numbers + Irrational numbers.
- Almost every number you use day to day is a real number.
- Numbers like $\sqrt{-1}$ are NOT real - they are called imaginary numbers.

7. Even and Odd Numbers

Definition Even numbers are exactly divisible by 2; odd numbers are not divisible by 2.

Examples: Even: 2, 4, 6, 8, 10 | Odd: 1, 3, 5, 7, 9

Notes:

- Even numbers end in 0, 2, 4, 6 or 8.
- Odd numbers end in 1, 3, 5, 7 or 9.
- Zero (0) is an even number.

8. Prime and Composite Numbers

Definition A prime number has exactly two factors (1 and itself). A composite number has more than two factors.

Examples: Prime: 2, 3, 5, 7, 11 | Composite: 4, 6, 8, 9, 10

Notes:

- 2 is the only even prime number.
- 1 is neither prime nor composite.
- Smallest prime is 2; smallest composite is 4.

Quick Revision Summary

The number families fit inside one another like this:

Natural < Whole < Integers < Rational < Real

Irrational numbers are also Real, but separate from Rational.

Real Numbers = Rational + Irrational (everything on the number line).